deskHPSDR TCI Command Support Matrix

Based on current upload: tci(106).c. Generated from the direct TCI dispatch table and checked against the initialization/broadcast code paths.

Summary

Area	Status
Direct inbound commands	58 dispatch entries
Protocol/device advertised	ExpertSDR3,2.0 / SunSDR2QRP
Advertised modes	LSB, USB, DSB, CW, FMN, AM, DIGU, SPEC, DIGL, SAM, DRM
DIGU/DIGL offsets	0..4000 Hz, sent before ready; owner-locked; mode-specific broadcast on mode change
Stream ownership	IQ stream owner lock; DIGU/DIGL offset owner lock

Notes

- Command names are case-normalized on receive.
- Commands listed in the first table are direct inbound dispatch entries in the current upload.
- DIGU/DIGL offsets are 0..4000 Hz and are owner-locked as one group, analogous to the IQ stream owner model.
- DIGU/DIGL offset affects RXA/TCI audio; DDC, TCI-IQ, VFO/CAT/QRG remain unshifted.
- On mode change, only the offset relevant to the new target mode is sent: DIGU -> digu_offset, DIGL -> digl_offset, other modes -> no DIG* offset.

Direct inbound TCI commands

These commands are present in the current tci_dispatch[] table and are therefore directly accepted by the server.

Command	Accepted form	Type	Implemented behavior	Notes
trx_count	trx_count;	Query	Returns number of TRX instances.	Startup also sends this.
trx	trx; / trx:<id>[,<true|false>[,...]];	Query/Set	Reports or requests TX/RX state. TX request goes through transmit-allowed/tune checks.	MOX/TRX broadcast maintained.
tune	tune; / tune:<id>[,<true|false>];	Query/Set	Reports or requests tune state.	Tune transition guarded.
split_enable	split_enable:<id>[,<bool>];	Query/Set	Reports or changes split state.	Uses current TX VFO logic.
lock	lock:<id>[,<bool>];	Query/Set	VFO lock alias/compatibility command.	Common lock handler.
vfo_lock	vfo_lock:<id>[,<vfo>,<bool>];	Query/Set	Reports or changes VFO lock.	Supports VFO-specific form.
sql_enable	sql_enable:<rx>[,<bool>];	Query/Set	Reports or changes squelch enable.	RX-scoped.
sql_level	sql_level:<rx>[,<level>];	Query/Set	Reports or changes squelch level.	RX-scoped.
rx_anf_enable	rx_anf_enable:<rx>[,<bool>];	Query/Set	Reports or changes ANF.	RX-scoped.
rx_apf_enable	rx_apf_enable:<rx>[,<bool>];	Query/Set	Reports or changes APF.	RX-scoped.
rx_nb_enable	rx_nb_enable:<rx>[,<bool>];	Query/Set	Reports or changes NB.	RX-scoped.
rx_nf_enable	rx_nf_enable:<rx>[,<bool>];	Query/Set	Reports or changes notch filter enable.	RX-scoped.
rx_bin_enable	rx_bin_enable:<rx>[,<bool>];	Query/Set	Reports or changes binaural mode.	RX-scoped.
rx_nr_enable	rx_nr_enable:<rx>[,<bool>];	Query/Set	Reports or changes noise reduction.	RX-scoped.
rit_enable	rit_enable:<vfo>[,<bool>];	Query/Set	Reports or changes RIT enable.	VFO-scoped.
xit_enable	xit_enable:<vfo>[,<bool>];	Query/Set	Reports or changes XIT enable.	TX/VFO logic.
rit_offset	rit_offset:<vfo>[,<Hz>];	Query/Set	Reports or changes RIT offset.	Broadcasts only when RIT enabled.
xit_offset	xit_offset:<vfo>[,<Hz>];	Query/Set	Reports or changes XIT offset.	Broadcasts only when XIT enabled.
digu_offset	digu_offset; / digu_offset:<0..4000>;	Query/Set	Reports or sets DIGU audio offset. Applies RXA shift and broadcasts.	Owner-locked: first setter owns DIGU/DIGL offset group.
digl_offset	digl_offset; / digl_offset:<0..4000>;	Query/Set	Reports or sets DIGL audio offset. Applies RXA shift and broadcasts.	Owner-locked; sent before ready and on DIGL mode change.
mute	mute; / mute:<bool>;	Query/Set	Reports or changes global mute.	Global.
rx_mute	rx_mute:<rx>[,<bool>];	Query/Set	Reports or changes RX mute.	RX-scoped.
volume	volume; / volume:<value>;	Query/Set	Reports or changes global volume.	Global audio.
rx_volume	rx_volume:<rx>,<ch>[,<value>];	Query/Set	Reports or changes RX channel volume.	Per RX/channel.
agc_gain	agc_gain:<rx>[,<value>];	Query/Set	Reports or changes AGC gain.	RX-scoped.
agc_mode	agc_mode:<rx>[,<mode>];	Query/Set	Reports or changes AGC mode.	RX-scoped.
iq_start	iq_start[:<rx>];	Start	Starts TCI IQ stream for receiver.	Global IQ sample-rate owner lock.
iq_stop	iq_stop[:<rx>];	Stop	Stops TCI IQ stream.	Owner release on stop/disconnect.
iq_samplerate	iq_samplerate:<rate>;	Set	Sets requested IQ sample rate.	Accepts supported rates; applied via GTK idle.
mon_volume	mon_volume; / mon_volume:<value>;	Query/Set	Monitor volume compatibility command.	Currently compatibility-level support.
mon_enable	mon_enable; / mon_enable:<bool>;	Query/Set	Monitor enable compatibility command.	Currently compatibility-level support.
audio_samplerate	audio_samplerate; / audio_samplerate:<...>;	Query/Set/Compat	Reports audio sample rate; accepts flexible argument forms.	TCI audio setup.
audio_stream_sample_type	audio_stream_sample_type; / ...	Query/Compat	Reports/accepts audio sample type negotiation.	Startup sends float32.
audio_stream_channels	audio_stream_channels; / ...	Query/Compat	Reports/accepts audio channel negotiation.	Startup sends 1 channel.
audio_stream_samples	audio_stream_samples; / ...	Query/Compat	Reports/accepts audio samples-per-frame negotiation.	TCI audio stream setup.
rx_sensors_enable	rx_sensors_enable:<rx>[,<ms>];	Set	Enables RX sensor/S-meter reporting.	Interval parsed; client flag stored.

Command	Accepted form	Type	Implemented behavior	Notes
tx_sensors_enable	tx_sensors_enable:<tx>[,<ms>];	Set	Enables TX sensor reporting.	Client flag stored.
spot	spot:<freq>,<callsign>,<mode>[,<...>];	Set	Adds/updates spot information.	Minimum 3, maximum 5 args.
spot_delete	spot_delete:<id>;	Set	Deletes a spot.	By identifier/argument.
spot_clear	spot_clear;	Set	Clears all spots.	No args.
audio_start	audio_start:<tx/rx>;	Start	Starts TCI audio. Supports RX audio and TX audio request path.	TX request activates TCI TX audio path.
audio_stop	audio_stop:<id>;	Stop	Stops TCI audio stream.	Clears stream state.
modulation	modulation:<vfo>[,<mode>];	Query/Set	Reports or changes mode.	Mode change broadcasts mode, filter band, and relevant DIGU/DIGL offset.
vfo	vfo:<id>,<channel>[,<Hz>];	Query/Set	Reports or changes VFO frequency.	VFO A/B mapping used.
rx_smeter	rx_smeter:<rx>[,<...>];	Query	Compatibility S-meter query.	Actual output uses rx_sensors form.
drive	drive; / drive:<tx>[,<value>];	Query/Set	Reports or changes drive.	Also updates tune drive if configured.
tune_drive	tune_drive:<tx>[,<value>];	Query/Set	Reports or changes tune drive.	Used for tune mode drive handling.
rx_filter_band	rx_filter_band:<rx>[,<low>,<high>];	Query/Set	Reports or changes RX filter band.	Normalizes/validates; ACK/broadcast path.
cw_macros	cw_macros:<idx>,<text>[,...];	Set	Defines/sends CW macro content.	CW engine integration.
cw_macros_stop	cw_macros_stop;	Set	Stops CW macro transmission.	CW engine abort/stop.
cw_msg	cw_msg:<text> / cw_msg:<idx>,<...>;	Set	Queues CW message/send request.	Supports multiple argument forms.
cw_terminal	cw_terminal:<bool>;	Set	Enables/disables CW terminal mode.	Broadcasts cw_terminal true/false.
cw_macros_speed	cw_macros_speed; / cw_macros_speed:<wpm>;	Query/Set	Reports or changes CW macro/keyer speed.	Also cw_keyer_speed alias present.
cw_keyer_speed	cw_keyer_speed; / cw_keyer_speed:<wpm>;	Query/Set	Reports or changes CW keyer speed.	Alias/compatibility.
cw_macros_delay	cw_macros_delay; / cw_macros_delay:<ms>;	Query/Set	Reports or changes CW macro delay.	CW macro timing.
cw_macros_speed_up	cw_macros_speed_up:<step>;	Set	Increases CW macro/keyer speed.	Step argument.
cw_macros_speed_down	cw_macros_speed_down:<step>;	Set	Decreases CW macro/keyer speed.	Step argument.
stop	stop;	Set	Stops client session reporting and marks client not running.	Clears sensor reporting.

Initial state and outbound/broadcast messages

Messages emitted during client initialization, query replies, or state broadcasts. Some are not direct inbound dispatch commands but are part of the TCI contract exposed by deskHPSDR.

Message / group	Form	When sent	Meaning
protocol	protocol:ExpertSDR3,2.0;	Initial state	Emulates ExpertSDR3 protocol level 2.0.
device	device:SunSDR2QRP;	Initial state	Advertised TCI device string.
receive_only	receive_only:true|false;	Initial state	Derived from transmitter availability / can_transmit.
trx_count	trx_count:<n>;	Initial state / query reply	Number of TRX instances.
channels_count	channels_count:2;	Initial state	Two-channel advertisement.
vfo_limits / if_limits	vfo_limits:<...>; if_limits:<...>;	Initial state	Sent by tci_send_limits().
iq_samplerate	iq_samplerate:<rate>;	Initial state / query reply	Current IQ sample rate.
audio_samplerate	audio_samplerate:<rate>;	Initial state / query reply	Current TCI audio sample rate.
audio_stream_sample_type	audio_stream_sample_type:float32;	Initial state	Float32 TCI audio.
audio_stream_channels	audio_stream_channels:1;	Initial state	Mono TCI audio stream.
modulations_list	modulations_list:LSB,USB,DSB,CW,FMN,AM,DIGU,SPEC,DIGL,SAM,DRM;	Initial state	Advertised modes.
dds / if / vfo / modulation	dds:...; if:...; vfo:...; modulation:...;	Initial and broadcast	Current VFO/DDS/mode state.
rx_filter_band	rx_filter_band:<rx>,<low>,<high>;	Initial and broadcast	Current RX filter band.
rx_enable / tx_enable	rx_enable:<rx>,true; tx_enable:...;	Initial state	RX/TX availability.
split_enable / lock / vfo_lock	split_enable:...; lock:...; vfo_lock:...;	Initial and broadcast	Split/lock state.
RX feature state	sql_enable, sql_level, rx_anf_enable, rx_apf_enable, rx_nb_enable, rx_nf_enable, rx_bin_enable, rx_nr_enable	Initial and broadcast	RX feature flags per receiver/VFO.
RIT/XIT state	rit_enable, rit_offset, xit_enable, xit_offset	Initial and broadcast	RIT/XIT state and offsets.
DIGU/DIGL offset	digu_offset:<n>; digl_offset:<n>;	Initial, set reply, broadcast	Sent before ready. Mode change sends only the target-mode offset.
TRX/Tune/Drive	trx, tune, drive, tune_drive	Initial and broadcast	TX state and drive settings.
Mute/Volume/AGC	mute, rx_mute, volume, rx_volume, agc_gain, agc_mode	Initial and broadcast	Audio and AGC state.
CW state	cw_macros_speed, cw_macros_delay, cw_keyer_speed, cw_terminal	Initial / broadcast	CW macro/keyer state.
ready / start	ready; start;	Initial state terminator	Sent after offsets and state are provided.